MFx – Macroeconomic Forecasting

Module: Introduction to EViews®

Note: This presentation serves as an outline of the topics discussed in the videos for this module.



IMFx

This training material is the property of the International Monetary Fund (IMF) and is intended for use in IMF Institute for Capacity Development (ICD) courses. Any reuse requires the permission of the ICD.
EViews® is a trademark of IHS Global Inc.

Objectives

- Examine the base dataset
- Understand the EViews interface
- Create and manage an EViews workfile
- Learn how to import data into EViews
- Forecast using a basic regression
- Learn how to use the model simulator
- Replicate your work using command scripts

Examine the Base Dataset

- Good practice to examine your base data before starting
- Helps to identify data issues, missing data, structural breaks
- Essential for verifying the data import step went well

EViews Interface



EViews Interface

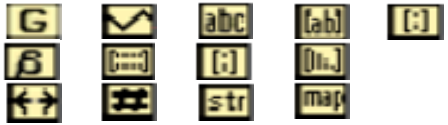
- Main Menu
- Command Window
- Object Window and Work Area
- Command Capture (EViews® 9)

EViews Objects



EViews Objects

- Yellow icons: data objects



- Blue icons: estimation objects



- Green icons: view objects



Five Key Objects

- Series 
- Groups 
- Samples 
- Equation 
- Model 

Workfiles, Series, and Groups



Workfiles, Series, and Groups

- Creating empty workfiles
  Pasting and importing data
- Creating a new series
  Series Object Menu
- Creating a new group
  Group Object Menu

Time-Series Data and Matrices



Time-Series Matrix

A data set with temporal ordering

Importing Data to EViews



Importing Data To EViews

- Copy and paste
- Importing data from foreign files (Microsoft Excel®)
- Drag and drop

Resid and c


Internationella valutafonden
국제통화기금
Fondo Monetario Internacional
Quỹ Tiền Tệ Quốc Tế
国際通貨基金
International Monetary Fund
國際貨币基金組织
Fondo Monetario Internazionale
صندوق النقد الدولي

`Resid` and `c`

- `resid`: An object for estimated residuals
- `c`: A vector containing the most recent estimates

The `smpl` Statement



The `smpl` Statement

- `smpl`: period used for estimations
- Controlling the active sample
- Ways to adjust the sample

Editing a Series



Editing a Series

- Changing data values and attributes
- Using interpolation to eliminate missing values

Summarizing Data



Summarizing Data

- Calculate basic statistics
- Different options to graph series

Creating Series from Existing Series



Generating Series from Existing Series

- Generate new series from existing series
- Basic transformations
- Calculating growth rates, taking logarithms

Working with Group Objects



Working with Group Objects

- Creating Group Objects
- Basic statistics for Group Objects
- Graphs

Estimating a Basic Regression



Estimating a Basic Regression

- The `ls` command
- Specifying the regression
- Setting the sample

Interpreting the results



Interpreting the results

- Parameter estimates
- Statistical significance
- Other statistical measures (R^2, SIC)

Allowing for Dynamics



Allowing for Dynamics

- Lagged dependent variables
- Lagged explanatory variables

Forecasting



Forecasting using the Forecast Tab

- Naming the forecast variable and standard errors
- Specifying the forecast sample
- Dynamic versus static forecasting

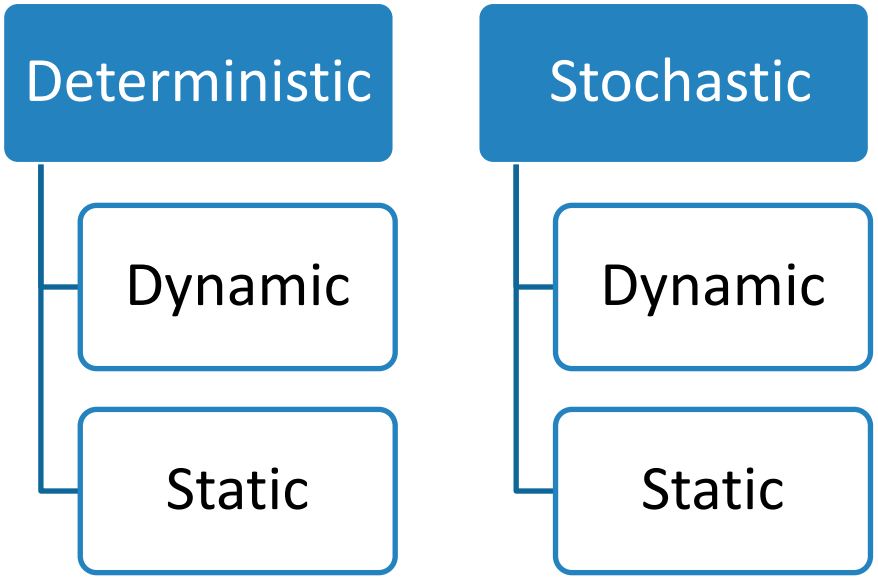
Forecasting Using the Model Simulator



Forecasting Using the Model Simulator

- Creating a model
- Solving the model

Deterministic versus Stochastic Forecasts



Forecasting when Assumptions Change



Forecasts when Assumptions Change

- Response of next year's real GDP in Fiji to a 10 percent rise/fall in sugar output.
- Another example: Response of inflation forecast to 100 basis point increase in the central bank lending rate

Replicating Your Work



Replicating Your Work

- Replication using script files
- Assembling commands in a script that can be executed in batch mode
- Running a program

Conclusion


Internationella valutafonden
국제통화기금
Fondo Monetario Internacional
Quỹ Tiền Tệ Quốc Tế
International Monetary Fund
国際通貨基金
صندوق النقد الدولي

Conclusion

- Importing data to EViews
- Manipulating series
- Representing series in graphs
- Creating groups
- Estimating and interpreting a basic regression
- Forecasting in different ways
- Replicating your work using script files